MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. The roots of $(x+1)(x-4)$ are
   - A. 1 and 4
   - B. -1 and -4
   - C. -1 and 4
   - D. 1 and -4
2. The discriminant of x^2-2x+3 equals
   - A. -8
   - B. 8
   - C. 16
   - D. -2
3. The vertex of $(x-2)^2+5$ is
   - A. $(2,-5)$
   - B. $(-2,5)$
   - C. $(5,2)$
   - D. $(2,5)$
4. The sum of the roots of $2x^2-6x+1$ equals
   - A. 6
   - B. 3
   - C. -3
   - D. $\frac{1}{2}$
5. $x^2-4<0$ is equivalent to
   - A. $x>2$
   - B. $x<-2$
   - C. $x\in(-2,2)$
   - D. $x\in[-2,2]$

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $x^2 + 6x + 9$ has as its factorisation
   - A. $(x-3)^2$
   - B. $(x+9)(x+1)$
   - C. $(x+6)^2$
   - D. $(x+3)^2$
2. The product of the roots of $3x^2 + x - 2$ equals
   - A. $-\frac{2}{3}$
   - B. $-\frac{1}{3}$
   - C. $\frac{2}{3}$
   - D. -2
3. $-x^2 + 4x$ reaches its maximum on $[0,1]$ at
   - A. 2
   - B. 0
   - C. 1
   - D. 4
4. A double root requires
   - A. $a = 0$
   - B. $\Delta = 0$
   - C. $b = 0$
   - D. $c = 0$
5. The polynomials of degree 2 with roots 0 and 2 are
   - A. $a(x+2)$
   - B. $a(x-2)^2$
   - C. $x(x-2)$ only
   - D. $ax(x-2)$, $a \neq 0$

True or false? Justify.

1. A quadratic function always has two real roots.
2. A quadratic with two distinct roots is negative between them.
3. For every real number x, $(x-2)^2 \geq 0$.

> **Key point · Self-assessment.** I justify the choice of form; I distinguish roots from the vertex; I express the solution of an inequality as a set; I respect the domain of the model.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.